

Determining Melting Curves Without Solid-Phase Information

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We report a robust dynamical signature of freezing in elemental liquids, revealed through an analysis of first-passage time properties of individual atomic motion. Specifically, we consider the mean first-passage time $\mathcal{T}(r)$ —the average time for an atom to reach a displacement r —and examine its behavior through the dimensionless logarithmic derivative $D(r) = d \ln \mathcal{T} / d \ln r$. In the stable liquid phase, $D(r)$ increases smoothly from 1 to 2, reflecting the expected crossover from ballistic to diffusive motion. As the freezing point is approached, this behavior changes qualitatively: $D(r)$ develops a distinct peak at a characteristic displacement r^* , reaching a threshold value $D(r^*) \approx 2.1$ that holds consistently across interaction potentials. This feature provides a robust and practical criterion for locating melting temperatures using molecular dynamics simulations of the liquid phase alone, without requiring information about the solid—particularly valuable when the solid phase is unknown or hard to determine. We demonstrate the method’s effectiveness in both model and real materials using *ab initio* molecular dynamics. Beyond its practical value, the emergence of this peak reflects the onset of dynamical hindrance, where atomic motion becomes sufficiently spatially localized to support the persistence of structural correlations and the emergence of crystallinity. Establishing a precise connection between the mechanistic criterion $D(r^*) \approx 2.1$ and the thermodynamic definition of freezing conditions remains an important direction for future work.

Accurate knowledge of the melting curve—the physical conditions defining solid–liquid phase equilibrium—is critical across a broad spectrum of scientific and technological domains, from production engineering [1] to astrophysics [2, 3]. In nuclear systems, for example, melting behavior governs the stability and performance of fuels and structural materials under extreme conditions [4, 5]. In geoscience, melting curves constrain models of planetary interiors, informing our understanding of core composition and thermal evolution [6]. Beyond their practical significance, melting curves reflect the profound—and sometimes exotic—changes in material bonding under pressure, phenomena now coming into experimental view [7]. Furthermore, from a fundamental perspective, the melting curve encodes the microscopic physics of the liquid–solid phase transition that, despite their apparent familiarity, remain rich with unresolved questions [8].

Despite steady progress, determining melting curves remains experimentally challenging, particularly under extreme conditions [9]. Theoretical predictions—especially from molecular simulations—therefore play a crucial supporting role [10]. However, all widely used computational methods—including free energy comparisons [11, 12], the Lindemann criterion [13, 14], two-phase coexistence [15], the Z-method [16], and heat-until-melting protocols [17]—rely on prior knowledge of the stable solid phase, a prerequisite that can be highly nontrivial. Under pressure, changes in electronic structure can challenge even first-principles predictions, often yielding multiple candidate phases and making identification of the stable solid phase difficult—even for elemental systems. A striking example is provided by the alkali metals, which evolve from prototypical simple metals into a series of complex, poorly understood structures, exhibiting nonmonotonic melting behavior and unusual electronic effects [7, 18–20].

A natural question therefore arises: can the melting curve be predicted without relying on prior knowledge of the solid phase? Several heuristic precursory criteria for freezing have been proposed, particularly for hard-sphere and Lennard-Jones liquid models [21–25], and colloidal systems [26]. Among these, the Hansen-Verlet rule [21] is perhaps the most broadly applicable across interaction potentials. Physically motivated by the dominance of short-range repulsion and packing effects near freezing, it states that a liquid freezes when the height of the first peak in the static structure factor $S(k)$ reaches a threshold value of approximately 2.85, as observed in hard-sphere crystallization. This threshold, however, exhibits variation across interaction potentials. For inverse-power-law interactions $1/r^n$, $S(k)_{\max}$ at freezing decreases from about 3.05 for $n = 12$ to 2.57 for $n = 1$ [27]. Moreover, structural changes at high pressure can markedly compromise the applicability of the Hansen-Verlet criterion (see sodium $S(k)$ in Fig. S1 of Ref. [28]).

A Dynamical Freezing Criterion. Motivated by these limitations, we introduce a robust dynamical signature of freezing in elemental liquids, which leads to a practical molecular dynamics method for determining the melting curve using only liquid-phase dynamics. The method complements widely used approaches by requiring no prior knowledge of the solid phase. It is simple to implement, compatible with short trajectories and small system sizes, and well suited to computationally intensive settings such as *ab initio* simulations or those leveraging modern machine-learned interatomic potentials [29].

Like the Hansen-Verlet rule, our method is based on a threshold criterion. But instead of relying on static structural features, it draws from the microscopic dynamics of atomic motion and shows greater robustness across different interaction potentials.

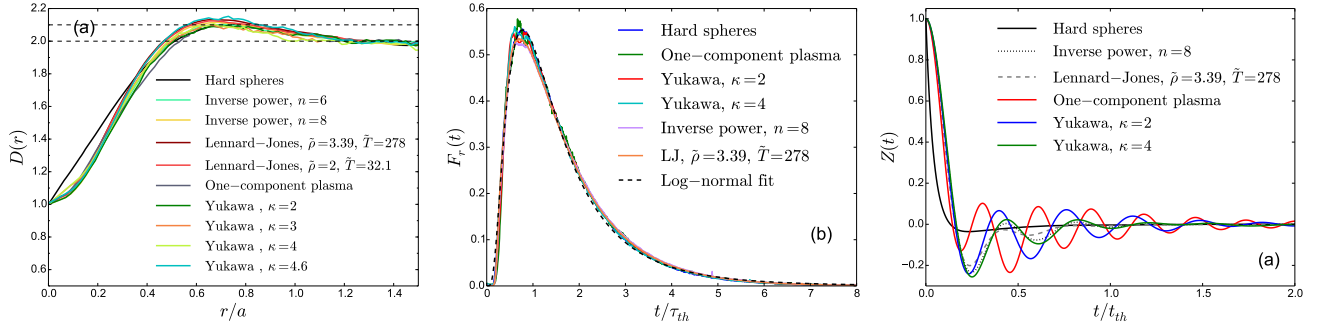


FIG. 1. (color online) (a) Logarithmic derivative $D(r)$ of the mean first-passage time for various 3D liquids at their freezing points. (b) First-exit time distributions $f(r, t)$ evaluated at the location $r = r^*$ of the peak in $D(r)$. The dashed line is a least-squares fit to a log-normal distribution. (c) Normalized velocity autocorrelation function $Z(t)$ for several 3D monatomic liquids at freezing. (In panels (b) and (c), a subset of cases from panel (a) is shown for clarity).

To identify this criterion, we adopt a different perspective on liquid dynamics. Traditional approaches track how observables such as the mean-square displacement or equilibrium time-correlation functions evolve with time. Here, we ask a complementary question: how long, on average, does it take for an observable to first reach a given value? This defines a first-passage problem. Borrowed from stochastic process theory [30], this perspective reveals dynamical signatures of freezing that are largely independent of microscopic details and elude traditional time-domain analyses.

Specifically, we consider the mean first-passage time $\mathcal{T}(r)$, defined as the average time it takes for an atom to reach a radial distance r from its initial position. From this, we analyze its logarithmic derivative,

$$D(r) = \frac{d \ln \mathcal{T}(r)}{d \ln r}, \quad (1)$$

which characterizes how the timescale for escape grows with radial displacement.

There are two key reasons for focusing on $D(r)$. First, as a dimensionless quantity, it enables direct comparison across systems with different interaction potentials, energy scales, or microscopic units. Second, its dependence on radial displacement reveals a clear and reproducible trend across systems. In a stable liquid above freezing, $D(r)$ increases monotonically from 1—reflecting inertial motion at infinitesimal timescales—to 2, consistent with diffusive behavior at long times. As the freezing point is approached—typically within 10% above the transition temperature—this smooth crossover breaks down: $D(r)$ develops a peak exceeding 2 at a characteristic displacement r^* , typically around 0.7 times the mean interparticle spacing a . As the temperature decreases further into the supercooled regime, the peak height $D(r^*)$ increases, while its location r^* shifts to smaller values (see Supplemental Material [31]).

The emergence of a peak in $D(r)$ signals the onset of *dynamical hindrance*—a qualitative shift in how particles

explore space, which we examine in more detail below. Most striking, however, is the consistency of this feature across diverse systems at the liquid–solid transition: in all cases studied, the peak converges to a well-defined value, $D(r^*) \approx 2.1$, at the transition temperature.

Figure 1a illustrates this behavior using molecular dynamics simulations of a broad range of model liquids at their respective freezing points (additional simulation details, interaction parameters, and system-specific results are provided in the Supplemental Material [31]). These include systems with hard-sphere, inverse-power-law, Lennard-Jones, Yukawa, and Coulomb interactions. Despite the wide variety of interaction potentials and thermodynamic conditions, all systems display the same characteristic pattern: $D(r)$ increases from 1, peaks near 2.1 at a characteristic displacement $r^* \sim 0.7a$, and then declines toward the diffusive limit of 2.

This level of agreement is particularly striking given that conventional dynamical observables at freezing—such as the velocity autocorrelation function (VAF)—depend sensitively on the form of the interparticle interaction. As shown in Fig. 1c, the VAF for the hard-sphere fluid decays rapidly and monotonically after a brief negative dip, consistent with short-lived, uncorrelated collisions. By contrast, the Lennard-Jones system exhibits sustained oscillations, reflecting transient caging. The Coulomb system also shows oscillations of similar origin, but with an added coupling to collective plasma modes; as a result, its deepest VAF minimum occurs at the second oscillation, rather than the first.

These differences are not merely quantitative—the VAF retains a strong imprint of interaction-specific dynamics and cannot be collapsed across systems by rescaling. In contrast, $D(r)$ shows a robust and reproducible form at freezing: it collapses onto a common shape and threshold value, $D(r^*) \approx 2.1$, regardless of interaction potential.

Remarkably, this universality extends beyond the average property $\mathcal{T}(r)$. The full first-passage time distribution $f(r, t)$, which gives the probability density for a

particle to first reach displacement r at time t and satisfies $\mathcal{T}(r) = \int_0^\infty t f(r, t) dt$, exhibits strong similarities across systems. As shown in Fig. 1, the distributions $f(r^*, t)$ collapse onto a single curve when time is scaled by the thermal timescale $t_{\text{th}} = a/v_{\text{th}}$, where a is the mean interparticle spacing and v_{th} is the thermal velocity. This collapse is observed consistently at the freezing point, indicating that the probability for a particle to reach this threshold in a scaled time t/t_{th} is nearly independent of the underlying interaction. Together, these results suggest that certain features of the freezing transition are largely insensitive to microscopic details—and are naturally revealed through first-passage statistics.

A Practical Criterion for Melting Curves. The consistent threshold value $D(r^*) \approx 2.1$ observed at freezing in simple model systems suggests a natural and practical application: using this feature as a criterion for determining melting curves. To explore this possibility, we applied the method to several elemental materials using *ab initio* quantum molecular dynamics (QMD) simulations based on density functional theory. In these simulations, ions evolve on the Born–Oppenheimer surface and interact through an effective potential that accounts for both direct ion–ion forces and the screening effects of conduction electrons.

We first considered aluminum and titanium as validation benchmarks, as both are well-characterized systems whose melting curves have been extensively studied using established simulation methods [32, 33]. Our QMD simulations employed comparable numerical setups [31], enabling direct comparisons with published results.

The numerical procedure was as follows. At a fixed density, we computed the pressure and the function $D(r)$ at various temperatures. The melting point was then identified as the temperature at which the maximum of $D(r)$ reaches the threshold value $D(r^*) = 2.1$. The resulting predictions, shown in Fig. 2 (top panel), are in close agreement with experimental reference values for both materials.

In addition to this agreement, our method compares favorably with two widely used approaches for melting curve prediction. The Z-method (blue triangles) estimates the melting point from the superheating limit of the solid, while the coexistence method (red diamonds) simulates the dynamics of liquid and solid phases in contact. In contrast, our method requires only liquid-phase data and makes no assumptions about the structure of the solid—an advantage in systems where the stable solid phase is unknown or difficult to determine. Moreover, our approach is computationally efficient: the results in Fig. 2 were obtained with systems of just $N = 64$ atoms, compared to $N = 256$ – 512 in previous studies [32, 33], and using trajectories only a few picoseconds long—sufficient for all atoms to reach r^* during the simulation.

We next applied the method to lithium, a system noted for its rich and anomalous high-pressure behavior [19].

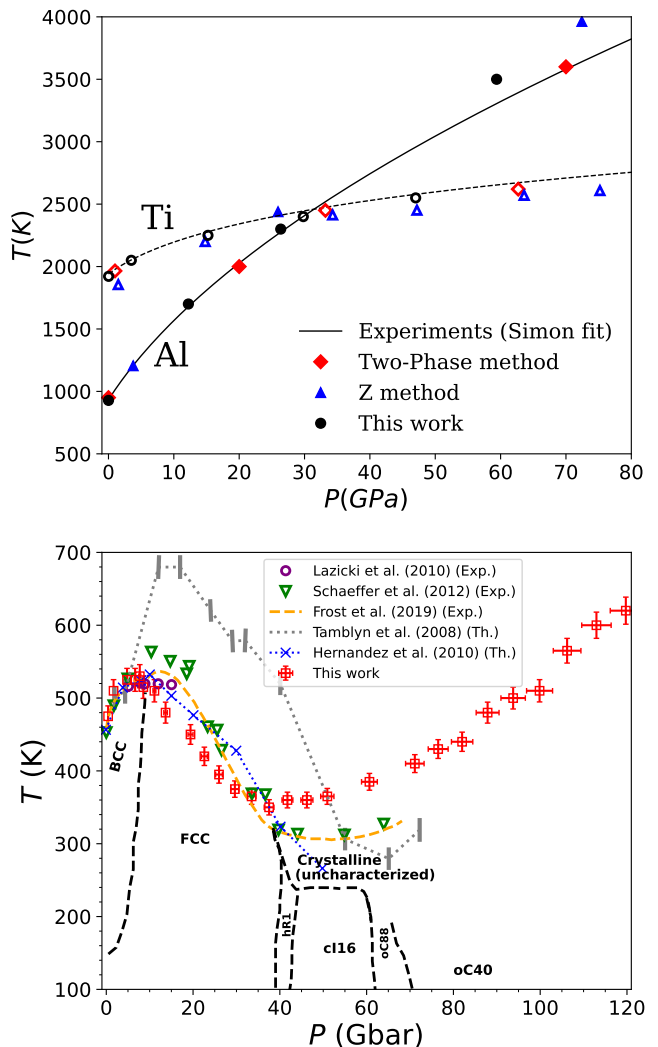


FIG. 2. (color online) (Top panel) Melting lines of aluminum (solid line and closed symbols) and titanium (dashed line and open symbols). Symbols correspond to QMD results using the methods shown in the legend; Z-method and coexistence data are from Refs. [32, 33]. Solid lines are Simon fits to experimental data from Ref. [34] for Al and Ref. [33] for Ti. (Bottom panel) Melting curve of lithium, reproducing the phase diagram from Ref. [19]. (Exp.) and (Th.) denote experimental and theoretical data from Refs. [19, 35, 36] and [37, 38], respectively. Tamblyn et al. used heat-until-melting; Hernandez et al. used two-phase coexistence.

Lithium exhibits a complex, nonmonotonic melting curve, with a reentrant regime in which the melting temperature decreases with increasing pressure, along with structural complexity and pressure-induced electronic transitions that pose challenges for both experiment and theory. Despite these complexities—and the known limitations of standard PBE GGA-based density functionals used here at high pressures [31]—our results shown in Fig. 2 (bottom panel) capture both the overall shape and the main trends of the melting curve, qualitatively and quantita-

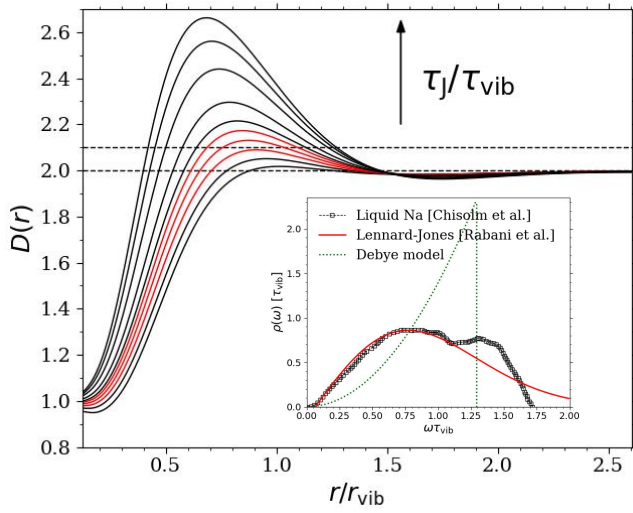


FIG. 3. (color online) CTRW model predictions of $D(r) = d \ln \mathcal{T}(r) / d \ln(r)$ for liquid sodium at various jump times $\tau_J/\tau_{\text{vib}} = 0.74, 0.86, 0.98, 1.1, 1.23, 1.48, 1.97, 2.46$, and 2.96 . Curves highlighted in red correspond to $\tau_J/\tau_{\text{vib}} = 0.86, 0.98, 1.11$. Inset: vibrational density of states $\rho(\omega)$ for liquid Na at freezing [39] (used in main panel) and for a Lennard-Jones liquid at freezing [40].

tively. This test highlights the method’s potential as a predictive tool, even in regimes where the solid phase is poorly characterized and the underlying electronic structure evolves in nontrivial ways.

Model-Based Physical Interpretation. We have now completed the presentation of the method. We find it interesting to close with a brief physical interpretation that offers intuition for the observed freezing signature. This perspective is not required to apply the method, but helps clarify the underlying mechanism. A full theoretical treatment will appear elsewhere; here, we focus on the essential ideas and implications (an outline is given in the Supplemental Material [31]).

To this end, we developed a minimal model inspired by the foundational ideas of Frenkel [41] and Zwanzig [42], who described atomic motion in liquids as a combination of localized vibrations and occasional spatial jumps—a perspective especially relevant near freezing. We describe this combined dynamics as a continuous-time random walk (CTRW) [43], which not only reproduces Zwanzig’s velocity autocorrelation function (VAF), but also provides a natural framework for computing first-passage statistics. Localized vibrations are characterized by a vibrational density of states $\rho(\omega)$, defining an average vibrational period $\tau_{\text{vib}} = 2\pi / \langle \omega \rangle$ and a root-mean-square displacement $r_{\text{vib}} = \sqrt{6k_B T \langle \omega^{-2} \rangle / m}$, where $\langle \cdot \rangle$ denotes averaging over $\rho(\omega)$. Spatial jumps occur with a mean waiting time τ_J , and their lengths are drawn from a distribution constrained by $\rho(\omega)$ and τ_J to match long-time transport properties, such as the self-diffusion coefficient. Thus, the

model depends only on $\rho(\omega)$ and τ_J .

To make the model concrete, we use the vibrational spectrum $\rho(\omega)$ obtained for liquid sodium near freezing shown in Fig. 3, based on quenched configurations and normal-mode analysis [39]. This leaves τ_J as the sole free parameter, serving as a proxy for temperature—larger values indicating lower temperatures. We evaluate $D(r)$ as a function of scaled distance r/r_{vib} for different values of τ_J . As shown in Fig. 3, the model captures the emergence and growth of the peak in $D(r)$ as τ_J increases, with the threshold $D(r^*) \approx 2.1$ reached when $\tau_J = \tau_{\text{vib}}$.

To test the robustness of this result, we repeated the analysis using a vibrational spectrum reported for Lennard-Jones liquids at freezing [40] (see Fig. 3). Despite differences at high frequencies, its low-frequency behavior closely matches that of sodium, and the resulting $D(r)$ curves show nearly identical peak height and position [31]. This indicates that the freezing signature in $D(r)$ is primarily controlled by low-frequency vibrational modes governing long-time escape dynamics.

These results support the interpretation that the peak in $D(r)$ marks the onset of dynamical hindrance—a transition from diffusive to transiently localized motion. While the precise relationship between the threshold $D(r^*) \approx 2.1$ and the thermodynamic freezing point remains to be established, the model suggests that this value corresponds to a critical degree of localization, beyond which structural correlations persist and crystalline order can emerge. Above freezing, atomic motion remains delocalized enough to erase structure; below it, caging and confinement stabilize local order.

We have identified a robust and reproducible dynamical signature of freezing, characterized by a peak in the logarithmic derivative $D(r)$ of the mean first-passage time. This feature, consistent across interaction potentials, enables melting curves to be determined from liquid-phase simulations alone, without requiring prior knowledge of the solid phase. The threshold value $D(r^*) \approx 2.1$ is concomitant with the onset of dynamical hindrance in atomic motion—from unhindered drifting motion to temporary spatial confinement—and invites further exploration of its connection to thermodynamic coexistence. A forthcoming publication will provide a detailed account of the method’s implementation and validation, comparisons with existing approaches, additional examples, and broader applicability.

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